Resolved referee R042 [Gap; revised] – The compactified closed-graph criterion.
Original source line 1117. The parenthesis appears intended to impose the correct compactified-target hypothesis; make it explicit by taking B to be a closed definable subset of $(\mathbb{P}^1(\mathrm{VF}))^m$, and take graph closure in $A \times B$. The example $f(0) = 0$, $f(x) = 1/x$ for $x \neq 0$ has closed graph in affine VF^2 but is discontinuous at 0: this is a counterexample only to the unbounded affine-target version, not to the intended compactified statement. For the reverse implication, apply curve selection to the set of input points x with $f(x) \notin \mathfrak{b}$, approaching a; apply the finite-limit lemma to the resulting finite-valued image in B. A limit point b' lies outside the clopen neighborhood $\mathfrak{b}$, and (a, b') lies in the graph closure. Also, "projective line with the valuation topology" is more precise than a literal topological one-point compactification; local compactness is not assumed here.

Author resolution. The author approves making the intended projective ambient space explicit: the affine target is B, and graph closure is in $A \times B^{\mathrm{P}}$, with the valuation topology. The reverse proof selects bad inputs first and then takes a finite projective limit of their images. Temporary base enlargement supplies curve selection without changing the geometric conclusion or imposing global VF-generation.

Unapplied R043/R044 proof proposal for author review. Both lemma statements are unchanged. In Lemma 2.51, the existing literal subsets of VF are already unary. The expanded proof explains the singleton limit and the equality of whole thin annuli using mutually inverse positive rational-affine laws of tail radii, without requiring slope one. In Lemma 2.53, the lower bound is direct; the upper bound uses justified finite nonempty choices over a temporary base and their nonempty minimal limit. Definable fiber endpoints provide one uniform bound in the RV-indexed case, by the same supremum method as the author's Lemma 2.41 induction. All green changes are unapplied, and the referee annotations remain unresolved pending review.

Lemma 2.51. *Let $f : A \longrightarrow B$ be a definable bijection between two open subsets of* VF*. Suppose that B is bounded. Then f is weakly concentric, that is, for every $a \in A$ there is a $b \in \mathrm{VF}$ such that, for every sufficiently large $\gamma \in \Gamma$, $f(\mathfrak{o}(a, \gamma) \smallsetminus a) = \mathfrak{o}(b, \delta) \smallsetminus b$ for some $\delta \in \Gamma$.*

Note that, compared to Lemma 2.32, the assumption of this lemma is stronger and its conclusion weaker, but the point is that we do not need a partition to achieve it.

Proof. Fix $a \in A$. Here A and B are unary, as in the statement. After translating the input and restricting to a small punctured disc about a, Lemma 2.45, applied with the closed bounded target $\overline{B}$, gives a finite minimal limit. Its cardinality is at most one by [17, Lemma 9.4]. It is nonempty because the function has a value at every point of the punctured disc. Write this limit as $\{b\}$.

By shrinking the punctured disc, exclude the at most one input for which $f(x) = b$. The valuation calculation in the proof of [17, Lemma 8.3], applied to the translated germ $f(x) - b$, gives a rational-affine law

$$\mathrm{val}(f(x) - b) = \rho(\mathrm{val}(x - a)), \qquad \rho(\gamma) = q\gamma + c,$$

for all sufficiently small nonzero $x - a$, where $q \in \mathbb{Q}_{>0}$ and $c \in \Gamma$. Indeed, unary polynomials have eventually monomial valuation, the definable germ selects one of the finitely many resulting valuation laws, and convergence to b forces positive slope.

By C-minimality, B contains a punctured neighborhood of b. On this neighborhood the inverse of f has a unique projective limit, using the projective finite-limit theorem and the same cardinality bound. That limit must be a, since $x \longrightarrow a$ implies $f(x) \longrightarrow b$ and the image avoids b on a sufficiently small punctured disc. Apply the same valuation calculation to the inverse to obtain a positive rational-affine law σ for its tail radii. Substituting $y = f(x)$ shows that $\sigma \circ \rho$ is the identity on a tail of Γ, so the two affine laws are inverse there.

Consequently, for every sufficiently large γ, with $\delta = \rho(\gamma)$, the image of the whole source thin annulus is exactly the whole target thin annulus:

$$f\big(\mathfrak{c}(a,\gamma) \setminus \mathfrak{o}(a,\gamma)\big) = \mathfrak{c}(b,\delta) \setminus \mathfrak{o}(b,\delta).$$

The forward inclusion is the law for f; the reverse inclusion follows by applying the inverse law to every point of the target annulus. Taking the unions over all radii greater than γ proves

$$f\big(\mathfrak{o}(a,\gamma) \setminus \{a\}\big) = \mathfrak{o}(b,\rho(\gamma)) \setminus \{b\},$$

as required. No assertion that $q = 1$ is needed. □

Referee R043 [Gap; revised] – The annulus equality needs a proof.
Original source line 1132. Specify that A and B each have one VF-coordinate. The finite limit is a singleton $\{b\}$ by [17, Lemma 9.4]; nonemptiness follows because the values are nonempty on a punctured neighborhood. The equality of whole thin annuli needs more explanation. Here is a repair. The valuation calculation in the proof of [17, Lemma 8.3], applied to $x \mapsto \mathrm{val}(f(x) - b)$, gives

$$\mathrm{val}(f(x) - b) = q\,\mathrm{val}(x - a) + c \qquad (q \in \mathbb{Q}_{>0})$$

on a sufficiently small punctured neighborhood: each unary polynomial has eventually monomial valuation, and the limit forces positive slope. Remove the at most one exceptional input with $f(x) = b$. By C-minimality, B contains a punctured neighborhood of b. The inverse has a unique projective limit there, necessarily a, since points $x \longrightarrow a$ give $f(x) \longrightarrow b$. Apply the same valuation calculation to the inverse. The resulting two affine maps of tail radii are inverse, so every sufficiently small target shell consists exactly of images of its corresponding source shell. Taking their unions proves the asserted punctured-disc equality. Thus the lemma is repairable; an additional assertion that $q = 1$ is unnecessary.

Definition 2.52. Let $p : A \longrightarrow \Gamma$ be a definable function. We say that p is an *$\mathfrak{o}$-partition* of A if, for every $a \in A$, the function p is constant on $\mathfrak{o}(a, p(a)) \cap A$.

This makes sense even when A has no VF-coordinates, in which case any definable function $A \longrightarrow \Gamma$ is an $\mathfrak{o}$-partition of A.

Lemma 2.53. *Let p be an $\mathfrak{o}$-partition of A. Suppose that A_{RV} is doubly bounded and A_t is closed and bounded for every $t \in A_{\mathrm{RV}}$. Then $p(A)$ is doubly bounded.*

Proof. First suppose that A has no RV-coordinates. The empty set is immediate. Choose $\eta \in \Gamma$ such that $A \subseteq \mathcal{O}^n_\eta$. If $p(a) < \eta$ for some $a \in A$, then $A \subseteq \mathfrak{o}(a, p(a))$, so the partition property makes p constant on A. Otherwise $p(A) \subseteq [\eta, \infty)$. In either case $p(A)$ is bounded below.

Suppose for contradiction that $p(A)$ is unbounded above. We may temporarily enlarge the parameter base to a small VF-generated base, as in Lemma 2.48; the desired bound is a geometric conclusion. For every $c \in \mathcal{M} \setminus \{0\}$ the definable set

$$A_{\mathrm{val}(c)} = \{x \in A : p(x) > \mathrm{val}(c)\}$$

is nonempty. Elementarity of the algebraic-closure model over this base together with c, followed by compactness, supplies a definable finitary function $F : \mathcal{M} \setminus \{0\} \longrightarrow \mathcal{P}(A)$ with finite nonempty values $F(c) \subseteq A_{\mathrm{val}(c)}$. Continuity of this choice is not needed.

By Lemma 2.45, F has a finite minimal limit set $L \subseteq A$, nonempty because its values are nonempty arbitrarily close to 0. Choose $e \in L$. Minimality implies that, for arbitrarily small nonzero c, $F(c)$ meets $\mathfrak{o}(e, p(e))$. Choose such a c with $\mathrm{val}(c) > p(e)$ and $x \in F(c) \cap \mathfrak{o}(e, p(e))$. The partition property at e gives $p(x) = p(e)$, whereas the choice gives $p(x) > \mathrm{val}(c) > p(e)$, a contradiction. This proves the upper bound.

For the general case, each nonempty fiber image $p(A_t)$ is bounded above and below by the case just proved, with t temporarily named. Definable completeness in Γ gives finite endpoints

$$\alpha_t = \inf p(A_t), \qquad \beta_t = \sup p(A_t).$$

Their defining greatest-lower-bound and least-upper-bound formulas make these functions definable over the original parameter base. Thus

$$t \longmapsto \gamma_t = \max\{|\alpha_t|, |\beta_t|, 0\}$$

is a definable Γ-valued function on A_{RV}. By Lemma 2.41, its image is doubly bounded. A common upper bound for the γ_t bounds $p(A)$ in both directions, proving the lemma. □

Referee R044 [Gap] – Bounds for an open-disc partition.
Original source line 1152. The lower bound has a short direct proof: choose η with $A \subseteq \mathcal{O}_\eta^n$. If $p(a) < \eta$ for some a, then $A \subseteq \mathfrak{o}(a, p(a))$, forcing p to be constant; otherwise $p(A) \subseteq [\eta, \infty)$. For the upper-bound argument, justify the nonempty finite choices in $A_{\mathrm{val}(c)}$ using the corrected curve-selection/model argument, and use the nonempty finite limit supplied by the limit lemma. In the general case, take a definable bound for each fiber image $p(A_t)$ by its definable endpoints in Γ, then apply Lemma 2.41; fiberwise boundedness by itself is not the uniform conclusion.

This lemma, in its various incarnations, is crucial for the good behavior of motivic Fourier transform (see [8, § 11] and [20]). For essentially the same reason, the main construction of this paper depends heavily on it (often in combination with Lemma 2.41).

Lemma 2.54. *Let $A \subseteq \mathrm{VF}^n$ be a definable set. Suppose that, for each $a \in A_1$, the fiber $A_a \subseteq \mathrm{VF}^{n-1}$ is open (respectively, closed). Then there is a finite definable set $A' \subseteq A_1$ such that $A \cap ((A_1 \smallsetminus A') \times \mathrm{VF}^{n-1})$ is open (respectively, closed) in $(A_1 \smallsetminus A') \times \mathrm{VF}^{n-1}$.*

Proof. Since either one of the two cases implies the other, we shall just concentrate on the open case. Consider the definable set

$$S = A \cap \mathrm{Cl}((A_1 \times \mathrm{VF}^{n-1}) \smallsetminus A),$$

where Cl is the closure operator in $A_1 \times \mathrm{VF}^{n-1}$; in other words, S is the boundary of A minus the frontier of A. It is enough to show that S_1 is finite. To that end, suppose for contradiction that S_1 contains a disc.

Since S cannot contain an open subset of VF^n, we have $\dim_{\mathrm{VF}}(S) < n$; for ease of notation, we may as well take $\dim_{\mathrm{VF}}(S) = n-1$. Then, up to a definable finite partition of S and possibly after a reordering of the last $n-1$ coordinates, we may assume that S is the graph of a definable continuous finitary function $\mathfrak{p} \longrightarrow \mathcal{P}(\mathrm{VF})$, where $\mathfrak{p}$ is an open polydisc and continuity is by Lemma 2.44. So, for all $\epsilon \in \Gamma$, if $\mathrm{rad}(\mathfrak{p})$ is sufficiently large, then S_n is clustered inside finitely many discs of radius ϵ. Now, for each $a \in S_1$, consider the function $\alpha(a, -) : S \longrightarrow \Gamma$ given by

$$\alpha(a, x) = \min\{\beta \in \Gamma \mid \mathfrak{o}(x, \beta) \subseteq A_a\},$$

which exists because A_a is open. By dimension theory, $\dim_{\mathrm{VF}}(\alpha^{-1}(\epsilon)) = n-1$ for some ϵ. Without loss of generality, we may assume that $\alpha(a, x) = \epsilon$ for all $(a, x) \in S$ and $\mathrm{rad}(\mathfrak{p})$ is sufficiently large in the sense described above. It follows that S contains an open polydisc of radius $\mathrm{rad}(\mathfrak{p})$, which contradicts the choice of S. □

Referee R045 [Gap; revised] – The fiberwise openness proof needs reconstruction.
Original source line 1173. The displayed proof contains genuine mistakes, but the conclusion has a repair. Retain $d = \dim_{\mathrm{VF}}(S) < n$ instead of replacing it by $n-1$. Define

$$\alpha(a, x) = \min\{\beta \geq 0 : \mathfrak{o}(x, \beta) \subseteq A_a\} \quad ((a, x) \in S).$$